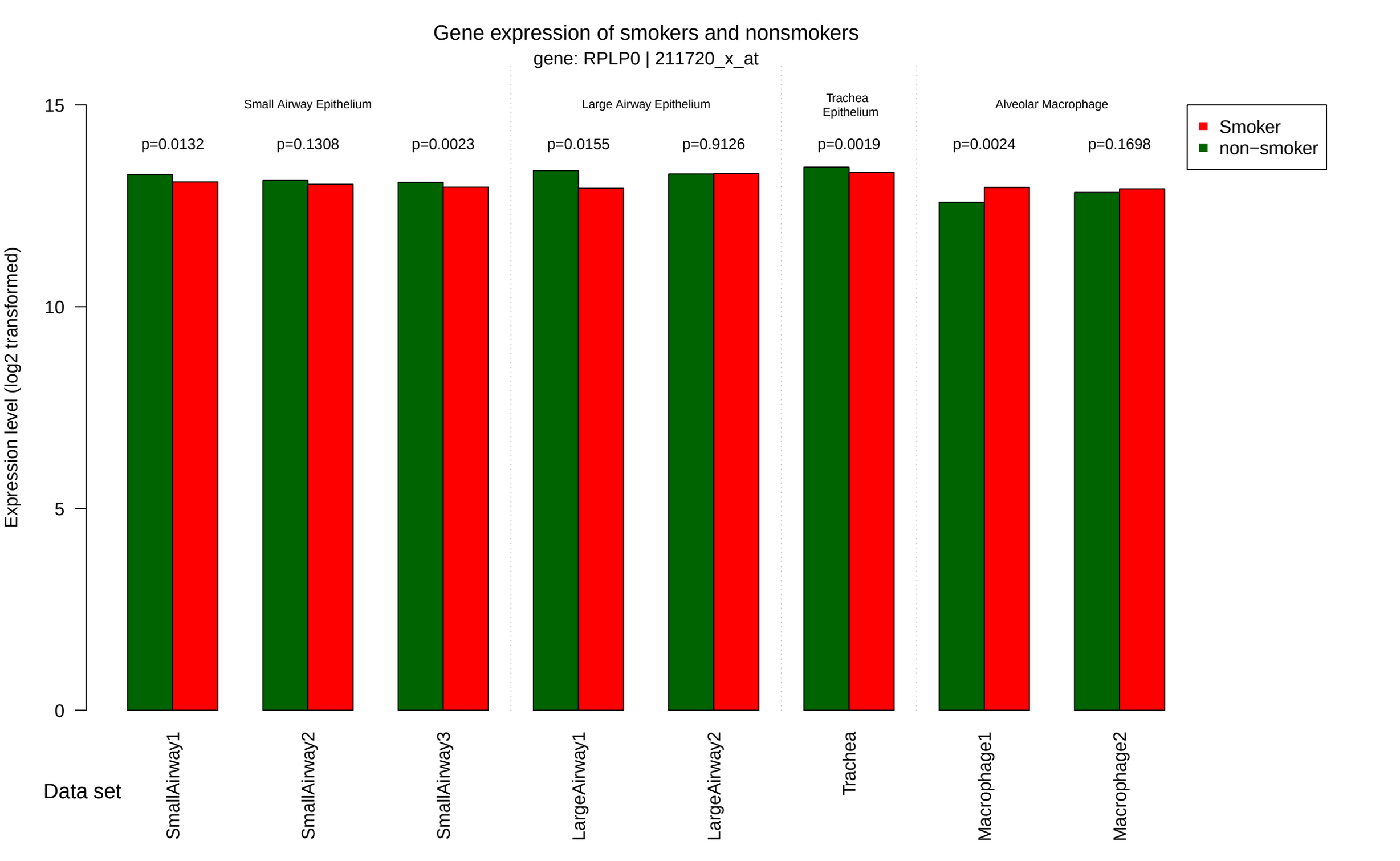
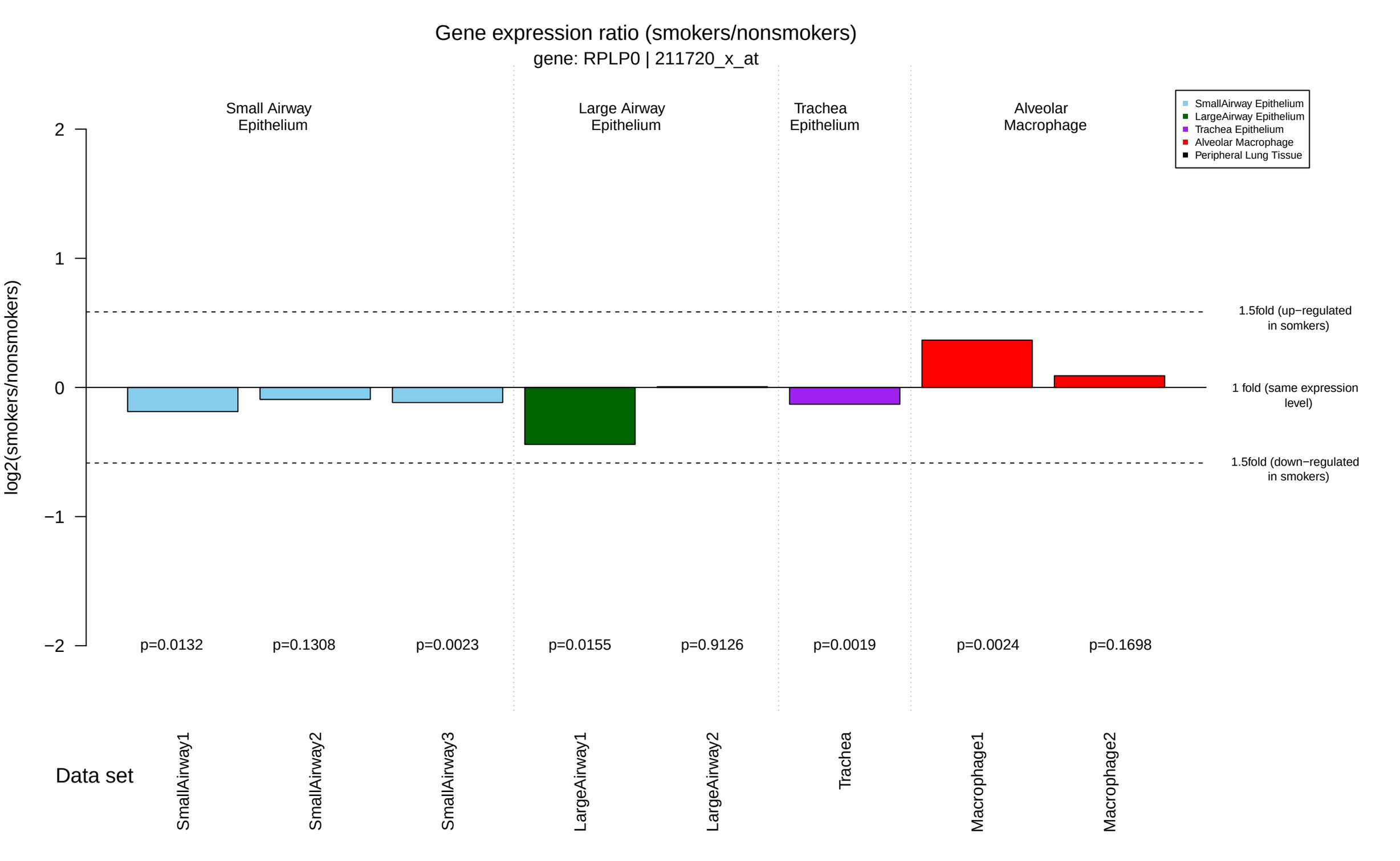
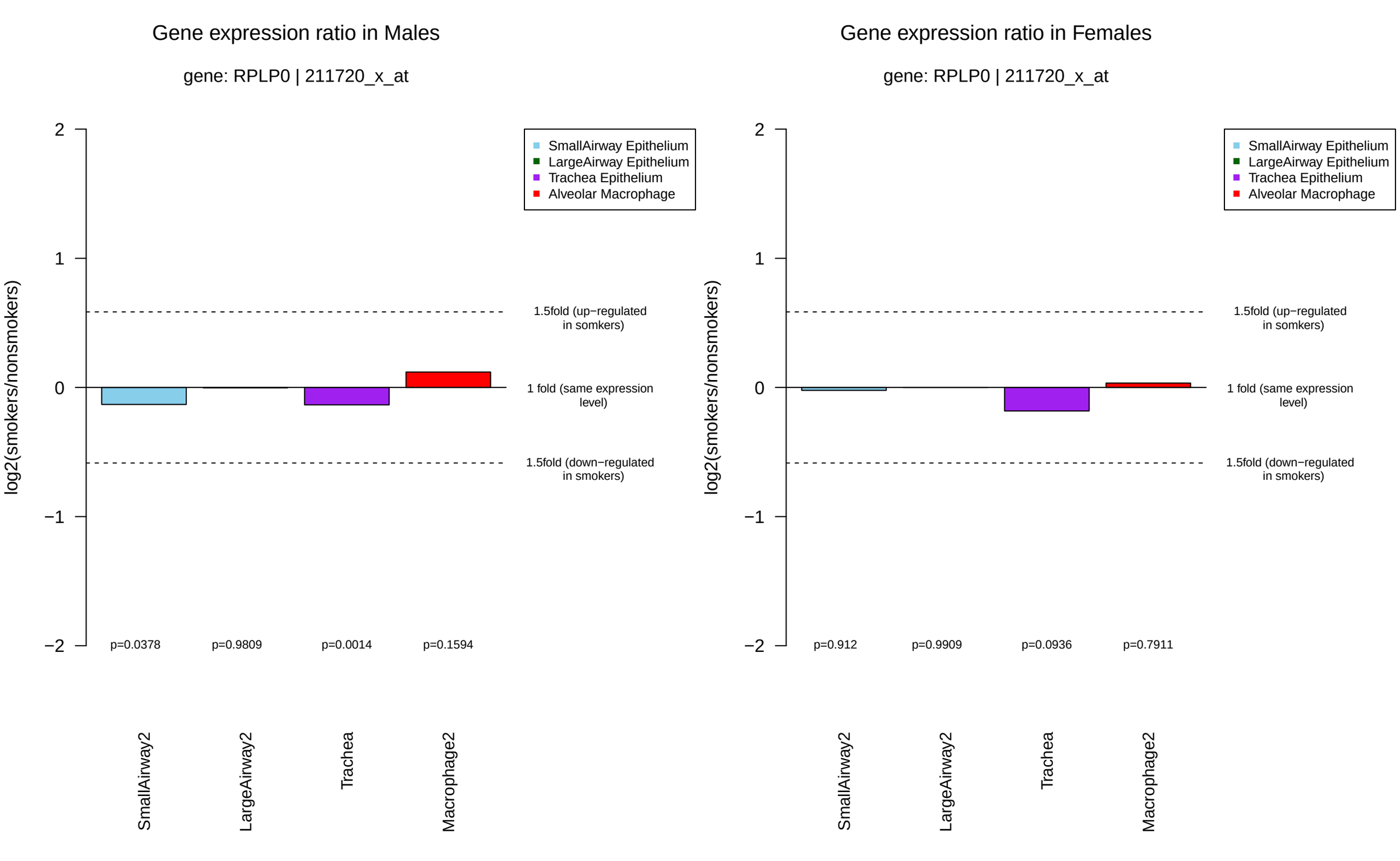
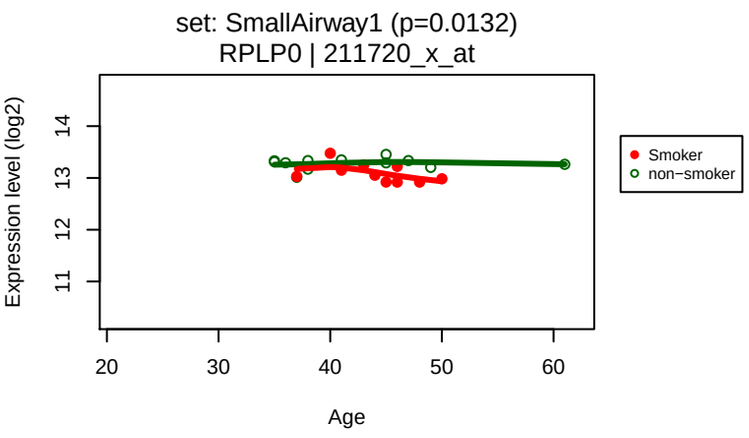
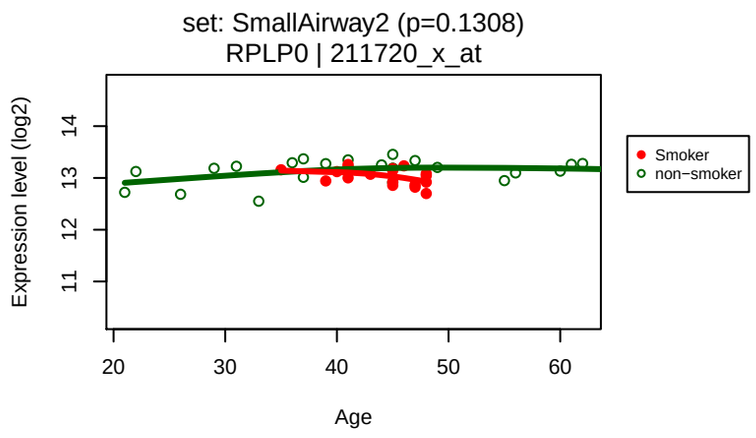
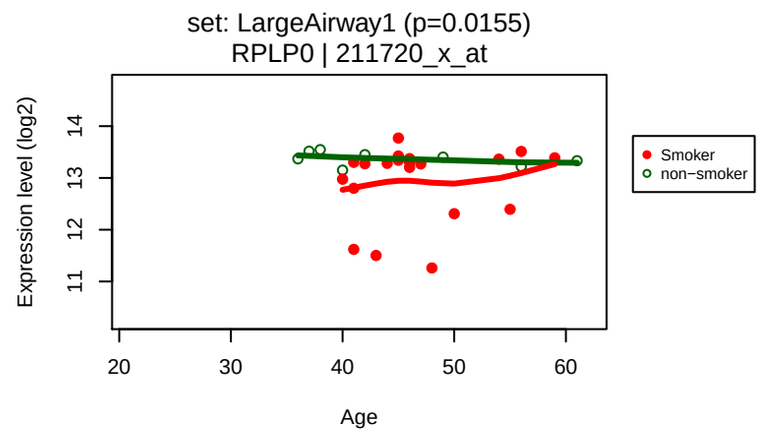

Gene expression of smokers and nonsmokers
gene: RPLP0 | 211720_x_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
p=0.0132
p=0.1308
p=0.0023
p=0.0155
p=0.9126
p=0.0019
p=0.0024
p=0.1698
Smoker
non−smoker
Expression level (log2 transformed)
15
10
5
0
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

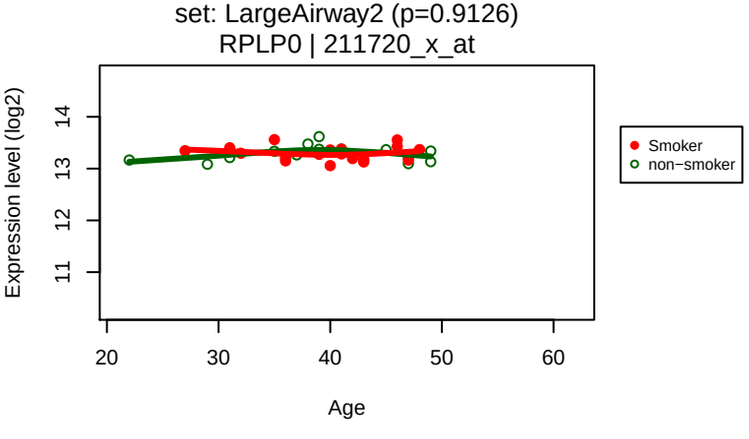

Gene expression ratio (smokers/nonsmokers)
gene: RPLP0 | 211720_x_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Peripheral Lung Tissue
log2(smokers/nonsmokers)
2
1
0
−1
−2
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.0132
p=0.1308
p=0.0023
p=0.0155
p=0.9126
p=0.0019
p=0.0024
p=0.1698
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

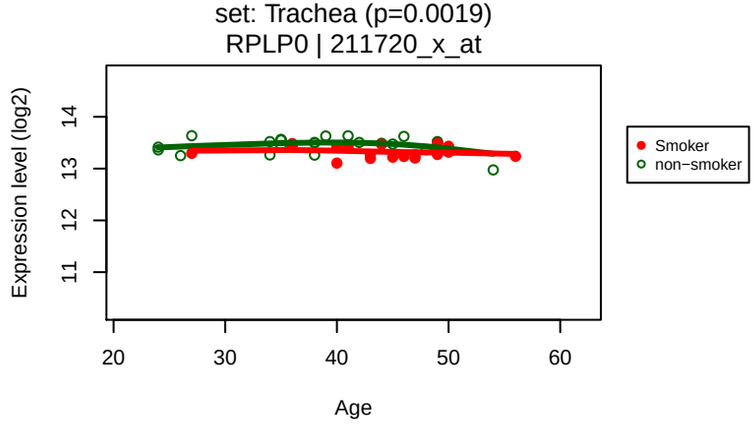

Gene expression ratio in Males
gene: RPLP0 | 211720_x_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.0378
p=0.9809
p=0.0014
p=0.1594
SmallAirway2
LargeAirway2
Trachea
Macrophage2
Gene expression ratio in Females
gene: RPLP0 | 211720_x_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.912
p=0.9909
p=0.0936
p=0.7911
SmallAirway2
LargeAirway2
Trachea
Macrophage2

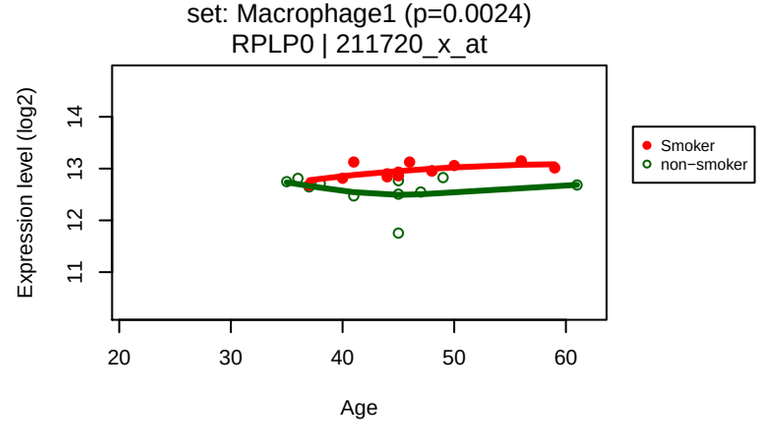

set: SmallAirway1 (p=0.0132)
RPLP0 | 211720_x_at
Expression level (log2)
Age
Smoker
non−smoker

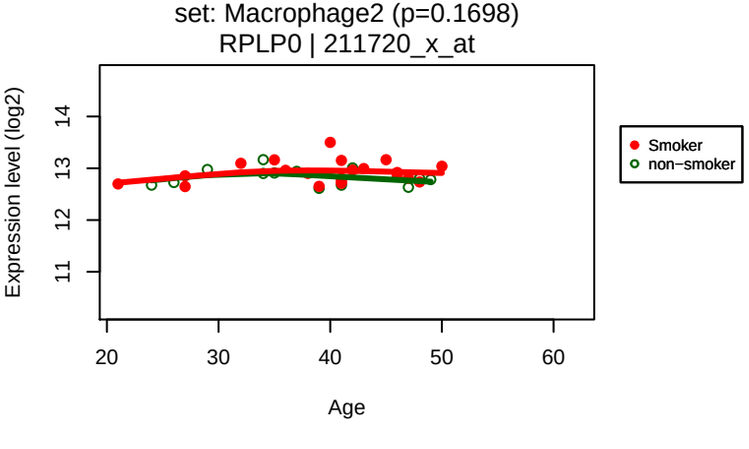

set: SmallAirway2 (p=0.1308)
RPLP0 | 211720_x_at
Expression level (log2)
Age
Smoker
non−smoker

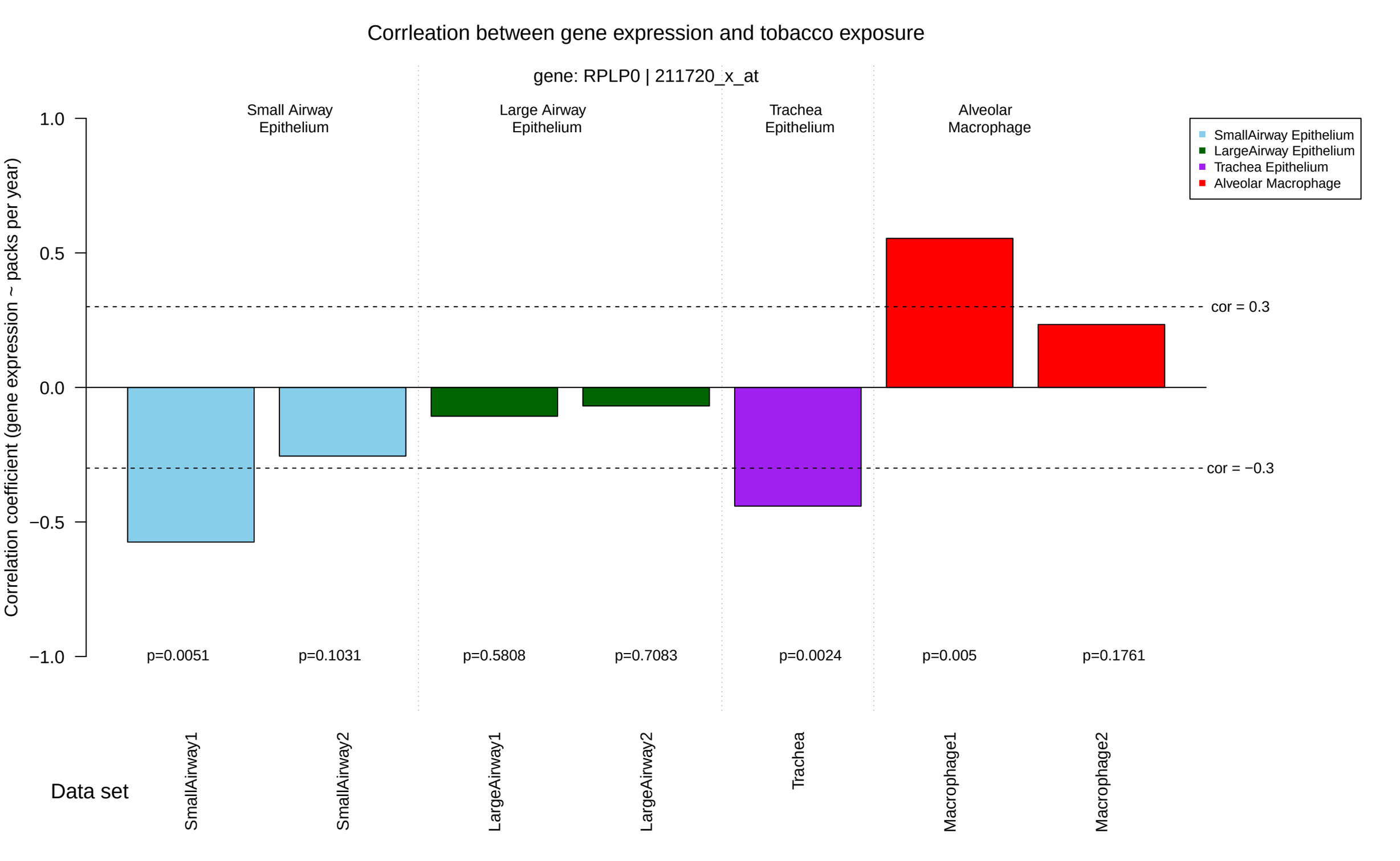

set: LargeAirway1 (p=0.0155)
RPLP0 | 211720_x_at
Expression level (log2)
Age
Smoker
non−smoker


set: LargeAirway2 (p=0.9126)
RPLP0 | 211720_x_at
Expression level (log2)
Age
Smoker
non−smoker


set: Trachea (p=0.0019)
RPLP0 | 211720_x_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage1 (p=0.0024)
RPLP0 | 211720_x_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage2 (p=0.1698)
RPLP0 | 211720_x_at
Expression level (log2)
Age
Smoker
non−smoker


Corrleation between gene expression and tobacco exposure
gene: RPLP0 | 211720_x_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Correlation coefficient (gene expression ~ packs per year)
1.0
0.5
0.0
−0.5
−1.0
cor = 0.3
cor = −0.3
p=0.0051
p=0.1031
p=0.5808
p=0.7083
p=0.0024
p=0.005
p=0.1761
Data set
SmallAirway1
SmallAirway2
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2